

I have a claim that made by a language model to a question, please help me for checking whether the claim can be entailed according to the provided reference which is related to the question.
The reference is a list of passages, and the claim is represented as a triplet formatted with ("subject", "predicate", "object").

If the claim is supported by ANY passage in the reference, answer 'Entailment'.
If NO passage in the reference entail the claim, and the claim is contradicted with some passage in the reference, answer 'Contradiction'.
If NO passage entail or contradict with claim, or DOES NOT contain information to verify the claim, answer 'Neutral'.

Please DO NOT use your own knowledge for the judgement, just compare the reference and the claim to get the answer.

Question:
{question}

Reference:
{reference}

Claim:
{claim}

Your answer should always be only a single word in ['Entailment', 'Neutral', 'Contradiction']. DO NOT add explanations or you own reasoning to the output.

Figure 16: Prompt for the GPT-4 Checker and Claude 2 Checker.

leading to the production of less reliable claim-triplets.

The outcomes of the Accurate Context are illustrated in Figure 21. Upon examination, it is evident that GPT-4-Turbo demonstrates the most notable performance, primarily generating responses aligned with the reference context. Conversely, GPT-3.5-Turbo tends to generate responses by relying on its internal knowledge to some extent, leading to contradictions or neutrality to the reference context. Claude 2 sometimes generate unsure information but neutral to the reference context. In the case of InstructGPT, the model further generates unsure information, which also contradicts the reference context. This behavior may stem from contradictions within the model’s internal knowledge or difficulties in comprehending the amalgamated content of internal and reference information. Regarding LLaMA-2-70B, and Falcon-40B-Instruct, our observations indicate that these models exhibit inferior performance. They generate information that contradicts internal knowledge and is irrele-

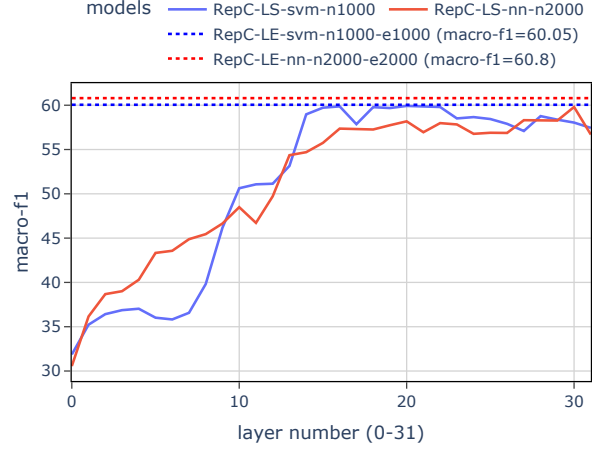


Figure 17: Performance tendency of different layers in RepC-LS checkers. The corresponding RepC-LE checkers are included as dashed lines.



Figure 18: Performance of different RepC checkers with respect to numbers of training samples.

vant to the reference context. Alpaca 7B performs similarly to GPT-3.5-Turbo, but seldom generates information contradicting to its internal knowledge. Different from the accurate context setting, all the models tend to generate more Neutral labels in the noisy context setting (Figure 22).

D.2 Internal Knowledge Bias of Checker

We also conduct an analysis to determine whether the checker provides predictions based on its internal knowledge. In this analysis, a triplet extracted

Triplet: ("The Story of June", "has songs", "Hey June")

Prompt: Translate the following subject-verb-object triplets into simple interrogative sentences: {Triplet}

Simple interrogative sentence: Does The Story of June has songs Hey June? Answer the question with Yes, No, or Unsure

Figure 19: Designed prompt for converting triplets to simple interrogative sentences.

Triplet: ("The Story of June", "has songs", "Hey June")
Question: What are the names of the songs from The Story of June which don't have Chinese names?
Prompt: Mask the objective or subjective in the subject-verb-object triplet { Triplet } in the Context with ##### which is not mentioned in Q (merely output the modified context). Q : { Question } Context: { Context }

Figure 20: Designed prompt for masking triplet information in the reference context.

Model	Checker	Entail	Contr	Neut
GPT-3.5	GPT-4	37.36	6.28	56.36
	Claude 2	61.49	25.95	12.56
	RoBERTa-NLI	21.82	21.76	62.64
GPT-4	GPT-4	35.88	10.88	53.24
	Claude 2	62.50	26.39	11.11
	RoBERTa-NLI	23.38	22.92	53.70

Table 14: Results for the information masking scenario in accurate-context setting.

from the response is taken, and we mask the subjective or objective information in the context with '#####'. The modified context, along with the triplet, is then inputted into the checker to obtain the label. In theory, the prediction label should be neutral because the relevant information in the context is masked. If the label is not neutral, it implies that the model is making inferences based on its internal knowledge. For the implementation of this analysis, we query GPT-4-Turbo with a specifically designed prompt to mask the triplet information, as illustrated in Figure 20. Specifically, in the noisy-context setting, we implement the query for each reference document and keep the document unchanged if there is no relevant information to the extracted triplet.

The results of the accurate context setting are shown in Table 14. As we observe, RoBERTa-NLI achieves the most significant Neutral labels, 62.64% and 53.70% for evaluated model GPT-3.5-Turbo and GPT-4-Turbo, respectively. The checker GPT-4-Turbo achieves the second performance. But Claude 2 predicts a large number of Entailment and Contradiction, which implies that the checker Claude 2 highly relies on the internal knowledge for checking. The results of the zero context setting are in a similar pattern with those of accurate-context setting (Table 15). But in the noisy context setting (Table 16), RoBERTa-NLI outperforms GPT-4-Turbo and Claude 2 with a large margin in the ratio of Neutral labels. The results may results from the strong bias to internal knowledge of

Model	Checker	Entail	Contr	Neut
GPT-3.5	GPT-4	37.91	22.55	39.54
	Claude 2	56.54	33.99	9.48
	RoBERTa-NLI	35.62	30.72	33.66
GPT-4	GPT-4	43.33	13.67	43.00
	Claude 2	61.00	28.00	11.00
	RoBERTa-NLI	34.67	23.00	42.33

Table 15: Results for the information masking scenario in zero-context setting.

Model	Checker	Entail	Contr	Neut
GPT-3.5	GPT-4	58.52	6.67	34.82
	Claude 2	60.25	20.00	19.75
	RoBERTa-NLI	9.38	10.62	80.00
GPT-4	GPT-4	65.71	6.29	28.00
	Claude 2	58.57	32.29	9.14
	RoBERTa-NLI	8.57	11.14	80.28

Table 16: Results for the information masking scenario in noisy-context setting.

GPT-4-Turbo and Claude 2 when the context is extremely long, or the RoBERTa-NLI model has less associative ability to the memorized knowledge.

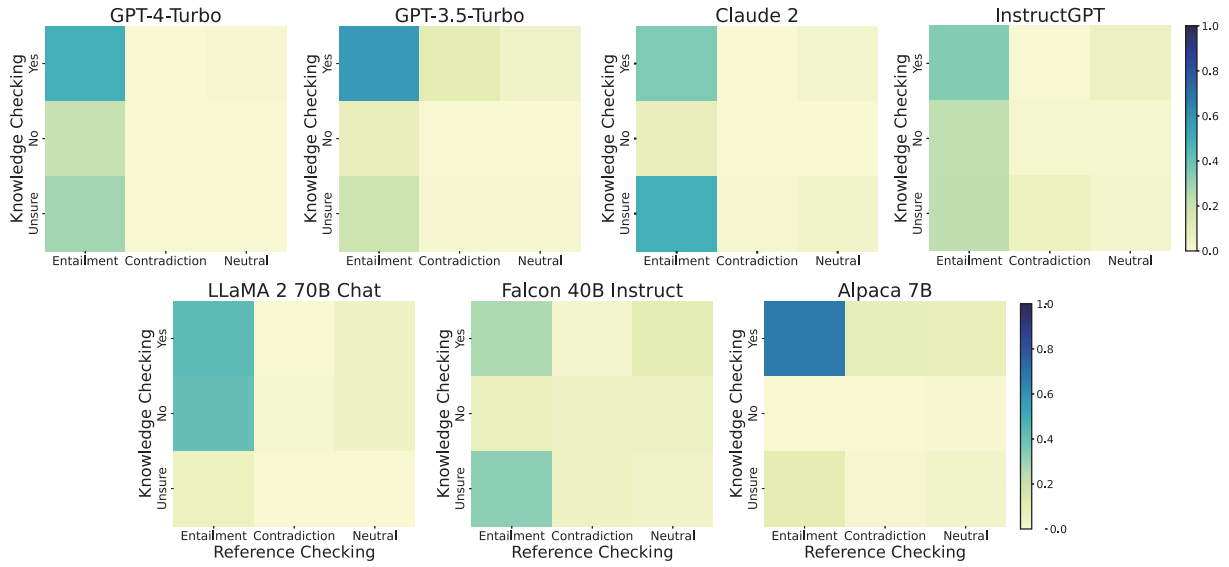


Figure 21: The results of knowledge checking for evaluated models in the accurate-context setting. The labels Yes, No and Unsure are the responses to the interrogative sentences generated from knowledge triplets. Each value refers to the percentage of each checking pairs in the total number of triplets.

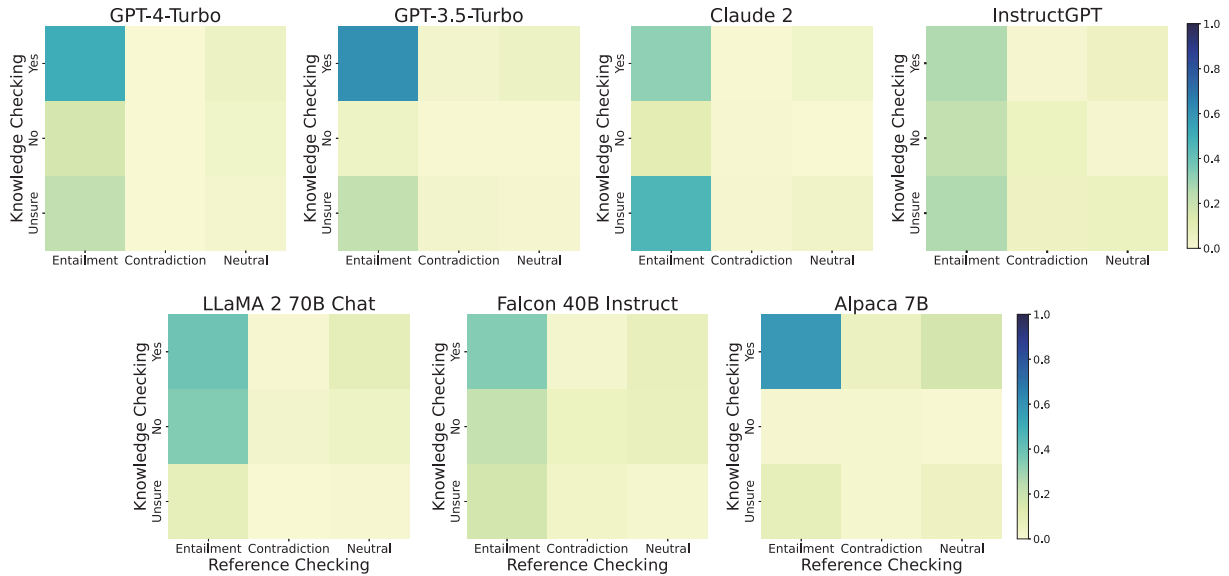


Figure 22: The results of knowledge checking for evaluated models in the noisy-context setting. The label Yes, No and Unsure are the responses to the interrogative sentences generated from knowledge triplets. Each value refers to the percentage of each checking pairs in the total number of triplets.